Problem sheet 10 for Rigid Analytic Geometry Summer Term 2026

For the first six of the eight problems from this sheep we keep the notations from the previous sheet (safe for the last problem of it). Thus K is complete with respect to the non-trivial ultrametric absolute value $|\cdot|$, $\mathbb{T}_1$ the one variable Tate algebra over K, $X = \mathrm{Sp}\mathbb{T}_1$ and X^* the space of van der Put points of X. In the following, a van der Put point is always a van der Put point of X.

Problem 1 (5 points)**.** *Let $r \in (0,1)_{\mathbb{R}} \cap |K^\times|$. Consider N_r as in (1) on sheet 9. Let $(f_i)_{i=0}^m \in \mathbb{T}_1^{m+1}$ such that $\mathbb{T}_1 = \langle f_0, \dots, f_m \rangle_{\mathbb{T}_1}$, and let $\Omega = \mathcal{R}_X(f_0|f_1, \dots, f_m) \subseteq K^\circ$. Show the following:*

- *If $N_r(f_i) \leq N_r(f_0)$ for $1 \leq i \leq m$, then there is $R \in (r,1]_{\mathbb{R}}$ such that $B_{\leq R}(0) \subseteq \Omega \cup B_{\leq r}(0)$.*
- *If there is $i \in [1,m]_{\mathbb{N}}$ such that $N_r(f_0) < N_r(f_i)$, then there is $R \in (r,1]_{\mathbb{R}}$ such that $\Omega \cap B_{\leq R}(0) \subseteq B_{\leq r}(0)$.*

Moreover, show that there is a unique van der Put point $\xi \in X^*$ such that $\xi \cap \mathfrak{Rat}_X$ is the set of Ω for which the first point in the above alternative occurs.

Problem 2 (2 points)**.** *Let $\xi \in X^*$ such that $r := R_\xi$ is an element of $(0,1)_{\mathbb{R}} \cap |K^\times|$, such that $0 \in B_{\leq R}(\xi)$ for all $R \in (r,1]_{\mathbb{R}}$ and such that $B_{\leq R}(0) \notin \xi$. Show that ξ coincides with the van der Put point from the previous problem!*

Remark 1. *Let ξ be a van der Put point satisfying:*

(1) $R_\xi \in (0,1)_{\mathbb{R}} \cap |K^\times|$
(2) $B = \bigcap_{r \in (R_\xi, 1]_{\mathbb{R}}} B_{\leq r}(\xi) \neq \emptyset$
(3) $B \notin \xi$.

Then $B \in \mathfrak{K}_{\leq R_\xi}$ and the result of translating ξ by any $x \in B$ is a van der Put point to which the previous problem can be applied. We have thus obtained a description of all van der Put points with the above three conditions.

It remains to study the case of a van der Put point ξ satisfying the first two of the above three conditions together with $B \in \xi$. An equivalent reformulation of this case is by the two conditions that $R_\xi \in |K^\times|$ and that $B_{\leq R_\xi}(\xi)$ exists. Unfortunately, this case splits into another two subcases, the last such split to occur in our investigation of van der Put points of X.

Let $B \in \mathfrak{K}_{\leq R}$ with $R \in (0,1]_{\mathbb{R}} \cap |K^\times|$. We put

$$\mathcal{S}(B) = \left\{ S_R(x) \;\middle|\; x \in B \right\}$$

where

$$S_R(x) = \big\{ y \in K \;\big|\; |x-y| = R \big\}.$$

It is easy to see that there is bijection between $\mathcal{S}(B)$ and the set of $< R$-balls $B_<$ such that $B_< \subseteq B$, sending such a ball to $S = B \setminus B_< \in \mathcal{S}(B)$ and sending $S \in \mathcal{S}(B)$ to $B_< = B \setminus S$.

Problem 3 (1 point)**.** *Let B be as above, and let $\xi \in X^*$ such that $B \in \xi$. Show that there is at most one $S \in \mathcal{S}(B)$ such that $S \notin \xi$.*

In the classification of van der Put point such that $R = R_\xi \in (0,1]_{\mathbb{R}} \cap |K^\times|$, it is necessary to distinguish the cases where such an S exists and where no such S exists. We start with the first case, for which we need some preparations.

Problem 4 (2 points)**.** *If $g \in \mathbb{T}_1$, show that*

$$K^\circ \setminus \big\{ x \in K^\circ \;\big|\; |g(x)| < \|g\,|\mathbb{T}_1\| \big\}$$

is the intersection of finitely many elements of $\mathcal{S}(K^\circ)$.

Remark 2. *If $f \in \mathbb{T}_1$ and $B = B_{\leq R}(y)$ with $y \in K^\circ$ and $R \in (0,1]_{\mathbb{R}} \cap |K^\times|$, one has $\rho \in K$ such that $|\rho| = R$. Then the previous result can be applied to*

$$g(x) = f(\rho \cdot x + y)$$

and shows that

$$B \setminus \big\{ x \in B \;\big|\; |f(x)| < n \big\}$$

is the intersection of finitely many elements of $\mathcal{S}(B)$, with

$$n = \|g\,|\mathbb{T}_1\| = \max_{x \in B} |f(x)|\,.$$

We put $\|f\|_B = n$. By the multiplicativity of $\|\cdot\,|\mathbb{T}_1\|$, or simply by the result just mentioned and the fact that $\mathcal{S}(B)$ is infinite, it follows that $\|\cdot\|_B$ is a valuation of $\mathbb{T}_1$ with values in $[0,\infty)_{\mathbb{R}}$.

Problem 5 (5 points)**.** *In the above situation, let $(f_i)_{i=0}^m \in \mathbb{T}_1^{m+1}$ such that $\mathbb{T}_1 = \langle f_0, \ldots, f_m \rangle_{\mathbb{T}_1}$, and let $\Omega = \mathcal{R}_X(f_0 | f_1, \ldots, f_m) \subseteq K^\circ$. Show the following:*

- *If $\|f_i\|_B \leq \|f_0\|_B$ for $1 \leq i \leq m$, then there is a finite subset $F \subseteq \mathcal{S}(B)$ such that*

$$\bigcap_{S \in F} S \subseteq \Omega.$$

- *If there is* $i \in [1, m]_{\mathbb{N}}$ *such that* $\|f_0\|_B < \|f_i\|_B$*, then there is a finite subset* $F \subseteq \mathcal{S}(B)$ *such that*

$$\Omega \cap \bigcap_{S \in F} S = \emptyset.$$

Moreover, show that there is a unique van der Put point $\xi_B \in X^*$ *such that* $\xi_B \cap \mathfrak{Rat}_X$ *is the set of* Ω *for which the first point in the above alternative occurs.*

Problem 6 (2 points)**.** *Let* $\xi \in X^*$ *such that* $R_\xi \in |K^\times|$*, that* $B = B_{\leq R_\xi}(\xi)$ *exists, and that* $\mathcal{S}(B) \subseteq \xi$*. Show that* $\xi = \xi_B$*!*

Problem 7 (3 points)**.** *Let* A *be a topological ring,* B *an adic* A*-algebra and* M *an adic* B*-module. Show that* M *is adic as an* A*-module!*

In particular, the composition of two adic morphisms of topological rings is adic.

In his 1993 paper "Continuous valuations", p. 459 before Lemma 1.8, Huber defines the notion of adicity for a morphism $A \xrightarrow{f} B$ only in the case of Huber rings. His condition is the existence of a pair of definition $(A^\sharp, I)$ for A and of a bounded open subring $B^\sharp \subseteq B$ such that $(B^\sharp, f(I) \cdot B^\sharp)$ is a pair of definition for the Huber ring B. Continuity of f is of course a consequence of this.

Problem 8 (3 points)**.** *Let* $A \xrightarrow{f} B$ *be a continuous homomorphism between Huber rings. Show that* f *is adic in the above sense if and only if it is adic in the sense explained in the lecture.*

Of the 23 points from this sheet, three are bonus points which do not count in the calculation of the 50%-threshold for passing the exercise module.

Solutions should be e-mailed to your tutor s45qhou at uni hyphen bonn dot de before Monday July 6. Good luck!